

Extracting Entities for an ERD

Here's a systematic approach to identify all possible entities from a project idea:

1. Start with a Project Description (Written or Mental)

Write down your project idea in plain English — a paragraph or two describing what the system does, who uses it, and what it manages.

2. Apply the "Noun Extraction" Method

Read through your description and **highlight every noun or noun phrase**. Each noun is a candidate entity.

Example — "An online bookstore where customers can browse books, place orders, make payments, and leave reviews. Admins manage inventory and track shipments."

Nouns extracted: Customer, Book, Order, Payment, Review, Admin, Inventory, Shipment

3. Filter the Nouns — 3 Categories

Not every noun becomes an entity:

Category	What it is	Action
Entity	A thing with multiple attributes worth storing	✓ Keep
Attribute	A property of another entity	✗ Demote to attribute
Relationship	Describes a connection between entities	✗ Becomes a relationship line

Example: "Book title" → attribute of Book, not its own entity.

4. Use the 5 Entity Discovery Lenses

Go beyond nouns by asking these questions:

-  **Who?** — Who are the actors/users of the system? → Customer, Admin, Driver, Vendor
 -  **What?** — What objects/things does the system manage? → Product, Order, Invoice, Document
 -  **Where?** — Are there locations or places? → Warehouse, Address, Branch, Store
 -  **When?** — Are there events or transactions recorded over time? → Order, Payment, Appointment, Session
 -  **Why/How?** — Are there concepts that describe rules, categories, or classifications? → Category, Role, Status, Permission
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5. Look for Junction/Associative Entities

These appear when you have **many-to-many relationships** that carry their own data:

- Student ↔ Course → Enrollment (with grade, date)
 - Doctor ↔ Patient → Appointment (with date, reason)
 - Order ↔ Product → OrderItem (with quantity, price)
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6. Validate Each Entity Against 3 Rules

An entity is valid if it:

1. **Has a unique identifier** — can you assign a primary key to it?
 2. **Has at least 2–3 meaningful attributes** — not just an ID
 3. **Has relationships** — it connects to at least one other entity
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7. Practical Checklist to Run Through

For any project, always check if these common entity groups apply:

- **Users/Roles** — who interacts with the system?
 - **Core domain objects** — the main "things" the system is about
 - **Transactions/Events** — things that happen over time
 - **Reference/Lookup data** — categories, types, statuses
 - **Locations** — physical or logical places
 - **Logs/History** — audit trails, history records
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Quick Summary — The Process

1. Write project description
 2. Extract all nouns
 3. Filter: entity vs attribute vs relationship
 4. Apply 5 lenses (Who, What, Where, When, How)
 5. Identify junction entities (M:N relationships)
 6. Validate each entity (PK + attributes + relationships)
 7. Draw the ERD
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